



GX-27X

Instruction Set

Documentation

Revision History		
Rev.	Summary of Change	Date
01	Original	2/3/10

Gilson, Inc. World Headquarters

3000 Parmenter Street | P.O. Box 620027 | Middleton, WI 53562-0027, USA | Tel: 608-836-1551 | Fax: 608-831-4451

Gilson S.A.S. | 19, avenue des Entrepreneurs | BP 145, F-95400 VILLIERS LE BEL, France

www.gilson.com | sales@gilson.com | service@gilson.com | training@gilson.com

Table of Contents

Instruction Set Command Report.....	2
Legend	2
System.....	3
GX-27X.....	5
GX Solvent System.....	12
GX Direct Injection Module	14
406 Single Syringe Pump	16
Non-Volatile Memory Map	19
GX-27X.....	19
GX Solvent System.....	20
GX Direct Injection Module	20
406 Single Syringe Pump	20
Error Messages.....	22
GX-27X.....	22
GX Solvent System.....	23
GX Direct Injection Module	23
406 Single Syringe Pump	24

Instruction Set Command Report

Legend

Command parameter headers are light blue.

Return parameter headers are light red.

Motion command headers are light yellow.

System

Device Name	System
Version	1.2.1.0
Unit ID	-1

Command Name	Notes	Parameters					
Get Instruction Set	Gets the instruction set from the connected instrument.						
Delete Instruction Set	Deletes the instruction set from the connected instrument.						
Renew Lease	Renews the lease with the connected instrument.						
Drop Lease	Drops the lease with the connected instrument.						
Get Firmware Version	Gets the Firmware version of the connected instrument.						
Set MAC Address	Sets the new MAC address for the connected instrument.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		MAC	string			-	
Kill	Disconnects the specified client socket connection.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		SocketHandle	string			-	
Reset	Resets the Ethernet module.						
Refresh Beacon	Requests beacon information from the connected instrument.						
Admin	Requests the connected instrument for Admin privileges. NOTE: Application with admin privileges can execute any command on the instrument.						
Get Last Response	Gets the last response sent from the instrument.						
Status Rate	Sets the status rate of the instrument to the specified interval.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Rate	number			-	
		Full Status Interval	number			-	

Command Name	Notes	Parameters					
Get Client Connections	Gets all the clients information connected to the instrument.						
Get Status							
Flash		<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		UnitId	number			-	
		Action	string			-	
		Data	string			-	
Set NVM	Sets the non-volatile memory value for the specified address. See table.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		UnitId	number			-	
		Address	number			-	
		Value	string			-	
Get NVM	Gets the non-volatile memory value for the specified address. See table.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		UnitId	number			-	
		Address	number			-	

GX-27X

Device Name	GX-27x
Version	1.0.12.0
Unit ID	35

Status Command	Frequency
Get Input Contacts	Sent once per command
Get Output Contacts	Sent once per command
Get Solenoid Contacts	Sent once per command
Get 24V Contacts	Sent once per command
Get Diverter Valve	Sent once per command
Get Error	Sent once per command
Get XY Position	Always on
Get Z Position	Always on

Command Name	Notes	Parameters						
Get Status	Get the status of the XYZ Motor, X Position, Y Position, Z Position, Error description if any.	<u>Name</u>	<u>Type</u>	<u>Notes</u>				
		MotorStatus	String	XYZ Motor status ('E' for Error, 'R' for Running (or homing), 'U' for Unpowered (also Unhomed), 'P' for Parked)				
		X Position	Number	X Position within 0-389				
		Y Position	Number	Y Position within 0-294				
		Z Position	Number	Z Position base on z-arm used				
24V Contacts	Turns output contacts on/off.	Error	String					
		<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>	
		24V Contact 1	OnOff	On	-	-	On/Off	
24V Contact 1	Turns output contact-1 on/off.	24V Contact 2	OnOff	On	-	-	On/Off	
		<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>	
		24V Contact 1	OnOff	On	-	-	On/Off	

Command Name	Notes	Parameters					
24V Contact 2	Turns output contact-2 on/off	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		24V Contact 2	OnOff	On		-	On/Off
Buffered	Legacy.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Command	String			-	
Clear Error	Clears error code.						
Diverter Valve		<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Valve State	OnOff	On		-	
Get 24V Contacts	Reads external power contacts status.	<u>Name</u>	<u>Type</u>	<u>Notes</u>			
		24V Contact 1	OnOff				
		24V Contact 2	OnOff				
Get Output and 24V Contacts	Reads relay output & external power status	<u>Name</u>	<u>Type</u>	<u>Notes</u>			
		Contact 1	OnOff	Relay output contact-1 status On/Off			
		Contact 2	OnOff	Relay output contact-2 status On/Off			
		24V Contact 1	OnOff	External 24V power contact status On/Off.			
		24V Contact 2	OnOff	External 24V power contact status On/Off			
Get Error	Gets error code. See table.	<u>Name</u>	<u>Type</u>	<u>Notes</u>			
		Error	String				
Get Diverter Valve	Returns On/Off status of Diverter	<u>Name</u>	<u>Type</u>	<u>Notes</u>			
		Valve	OnOff	On/Off			
Get Input Contacts	Gets status of input contacts.	<u>Name</u>	<u>Type</u>	<u>Notes</u>			
		Contact A	OnOff				
		Contact B	OnOff				
		Safety	OnOff				
Get Motor Status	Gets XYZ Motor status. 'E' for Error, 'R' for Running (or homing), 'U' for Unpowered (also Unhomed), 'P' for Parked.	<u>Name</u>	<u>Type</u>	<u>Notes</u>			
		X Motor	String				
		Y Motor	String				
		Z Motor	String				

Command Name	Notes	Parameters						
Get Output Contacts	Gets relay output contacts status(On/Off)	<u>Name</u>		<u>Type</u>		<u>Notes</u>		
		Contact 1		OnOff				
		Contact 2		OnOff				
Get Solenoid Contacts	Gets state of solenoid contacts.	<u>Name</u>		<u>Type</u>		<u>Notes</u>		
		Solenoid 1		OnOff				
		Solenoid 2		OnOff				
Get XY Position	Gets X & Y positions. X position ranges from 0-389. Y position ranges from 0-294.	<u>Name</u>		<u>Type</u>		<u>Notes</u>		
		X Position		Number				
		Y Position		Number				
Get Z Position	Gets Z-position.	<u>Name</u>		<u>Type</u>		<u>Notes</u>		
		Z Position		Number				
Home	Homes XYZ. Moves Z to the top of travel. XY to the park position. Any error state is cleared before homing starts.	<u>Name</u>						
		X Motor						
		Y Motor						
		Z Motor						
Identify	The PROM instrument name and version	<u>Name</u>		<u>Type</u>		<u>Notes</u>		
		Identity		String				
Immediate	Legacy	<u>Name</u>		<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Command		String	%		-	
		<u>Name</u>		<u>Type</u>	<u>Notes</u>			
		Imd Return		String				
Move X	Sets the new X position.	<u>Name</u>		<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		X Position		Number	0	mm	0 - 389	
		<u>Name</u>						
		X Motor						

Command Name	Notes	Parameters					
Move X with Speed	Sets the new X Position with the specified speed.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		X Position	Number	0	mm	0 - 389	
		X Speed	Number	350	mm/sec	0 - 350	
		<u>Name</u>					
		X Motor					
Move X with Speed and Current	Moves to new X-axis position with the specified speed & Drive level	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		X Position	Number	0	mm	0 - 389	
		X Speed	Number	350	mm/sec	0 - 350	
		X Current	Number	100	percentage	0 - 100%	Drive Level
		<u>Name</u>					
		X Motor					
Move XY	Sets the new X & Y position.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		X Position	Number	0	mm	0 - 389	
		Y Position	Number	0	mm	0 - 294	
		<u>Name</u>					
		X Motor					
		Y Motor					

Command Name	Notes	Parameters					
Move XY with Speed	Set the new X & Y position with the specified speeds	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		X Position	Number	0	mm	0 - 389	
		X Speed	Number	350	mm/sec	0 - 350	
		Y Position	Number	0	mm	0 - 294	
		Y Speed	Number	350	mm/sec	0 - 350	
		<u>Name</u>					
		X Motor					
		Y Motor					
Move XY with Speed and Current	Sets the new X & Y position with the specified speed and drive level.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		X Position	Number	0	mm	0 - 389	
		X Speed	Number	350	mm/sec	0 - 350	
		X Current	Number	100	percentage	0 - 100%	
		Y Position	Number	0	mm	0 - 294	
		Y Speed	Number	350	mm/sec	0 - 350	
		Y Current	Number	100	percentage	0 - 100%	
		<u>Name</u>					
		X Motor					
		Y Motor					
Move Y	Set the new Y position.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Y Position	Number	0	mm	0 - 294	
		<u>Name</u>					
		Y Motor					

Command Name	Notes	Parameters					
Move Y with Speed	Set the new Y position with the specified speed.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Y Position	Number	0	mm	0 - 294	
		Y Speed	Number	350	mm/sec	0 - 350	
		<u>Name</u>					
		Y Motor					
Move Y with Speed and Current	Set the new Y position with the specified speed and drive level.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Y Position	Number	0	mm	0 - 294	
		Y Speed	Number	350	mm/sec	0 - 350	
		Y Current	Number	100	percentage	0 - 100%	
		<u>Name</u>					
		Y Motor					
Move Z	Sets the new Z position.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Z Position	Number	0	mm	-	
		<u>Name</u>					
		Z Motor					
Move Z with Speed	Sets the new Z position with the specified speed.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Z Position	Number	0	mm	-	
		Z Speed	Number	125	mm/sec	0 - 125	
		<u>Name</u>					
		Z Motor					

Command Name	Notes	Parameters					
Move Z with LLD	Sets the new Z position with Liquid Level Detection. If liquid is encountered (on a downward movement) or air is encountered (on an upward movement) the movement stops immediately. If the position is the same as the target specified it may be assumed that liquid (or air) was not encountered	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Z Position	Number	0	mm	-	Range is based on the Z-Arm used.
		Z Speed	Number	40	mm/sec	0 - 40	
Output Contacts	Set the relay output contacts to On/Off	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Contact 1	OnOff	On		-	
Output Contact 1	Sets the relay output contact-1 to On/Off.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Contact	OnOff	On		-	
Output Contact 2	Sets the relay output contact to On/Off.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Contact	OnOff	On		-	
Pulse Contact	Sets the specified relay output contact for pulse time specified. For example values 1 and 0.5 sets the relay contact 1 to be closed for 0.5 seconds.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Contact	Number	1		-	Relay output contact number(1 or 2)
		Time	Number	0.1	secs	-	
Solenoid Outputs	Sets the state of the solenoid outputs.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Solenoid 1	OnOff	On		-	
		Solenoid 2	OnOff	On		-	
Solenoid 1	Sets the state of the Solenoid output 1	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Solenoid	OnOff	On		-	
Solenoid 2	Sets the state of the solenoid output 2	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Solenoid	OnOff	On		-	
Wait For X	Wait for X-Motor to come to parked position.	<u>Name</u>					
		X Motor					

Command Name	Notes	Parameters
Wait For XY	Wait until X & Y Motors to come to parked position.	<u>Name</u> X Motor Y Motor
Wait For Y	Wait for Y-Motor to come to parked position.	<u>Name</u> Y Motor
Wait For Z	Wait for Z-Motor to come to parked position.	<u>Name</u> Z Motor

GX Solvent System

Device Name	GX Solvent System
Version	1.0.8.0
Unit ID	8

Status Command	Frequency
Get Error	Sent once per command
Get Syringe Info	Always on

Command Name	Notes	Parameters					
Aspirate	Flip valve to position specified and then aspirate specified volume with the specified flow rate.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Valve Position	String	1		-	'N' for needle 'R' for reservoir
		Volume	Number	0	μL	-	Volume to aspirate
		Flow Rate	Number	10	mL/min	-	Flow rate in mL/min at which to aspirate
		<u>Name</u>					
		Syringe					
Buffered	Legacy.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Command	String			-	
Clear Error	Clears error code.						

Command Name	Notes	Parameters					
Dispense	Flip valve to position and dispense the volume specified.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Valve Position	String	1		-	'N' for needle 'R' for reservoir
		Volume	Number	0	μL	1 - 50000	Volume in μL to dispense
		Flow Rate	Number	10	mL/min	-	Flow rate in mL/min at which to dispense
		<u>Name</u>					
		Syringe					
Get Error	Gets error code. See table.	<u>Name</u>		<u>Type</u>	<u>Notes</u>		
		Error		String			
Get Motor Status	Gets motor status	<u>Name</u>		<u>Type</u>	<u>Notes</u>		
		Valve		String	'P' = parked, 'R' = running, 'E' =error condition, 'I' = not initialized		
		Syringe		String	'P' = parked, 'R' = running, 'E' =error condition, 'I' = not initialized		
Get Syringe Info	Gets the Valve position of the syringe and the Volume of the syringe contents. Default unit for volume is μL.	<u>Name</u>		<u>Type</u>	<u>Notes</u>		
		Valve Position		String	Valve position can be 'R' for Reservoir, 'N' for Needle, 'O' for valve motor overload, 'X' for running		
		Volume		Number	Volume can be to 0.1 μL resolution, depending on syringe		
Halt Syringe	Halt the syringe and generate error.	<u>Name</u>					
		Syringe					
Home	Homes GX Solvent system selection valve to waste position. The held volume is set to zero but no pumping occurs.	<u>Name</u>					
		Syringe					

Command Name	Notes	Parameters					
Identify	The PROM instrument name and version.	<u>Name</u>	<u>Type</u>	<u>Notes</u>			
		Identity	String				
Immediate	Legacy.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Command	String	%		-	
		<u>Name</u>	<u>Type</u>	<u>Notes</u>			
		Response	String				
Valve Position	Sets the valve position.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Valve Position	String	1		1 - 6	
		<u>Name</u>					
		Syringe					
Wait For Syringe	Wait for syringe motor to get into parked position.	<u>Name</u>					
		Syringe					

GX Direct Injection Module

Device Name	GX Direct Inject
Version	1.0.5.0
Unit ID	9

Status Command	Frequency
Get Error	Sent once per command
Get Valve Position	Sent once per command

Command Name	Notes	Parameters					
Buffered	Legacy.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Command	String			-	
Clear Error	Clears error code.						
Get Error	Gets error code. See table.	<u>Name</u>	<u>Type</u>	<u>Notes</u>			
		Error	String				

Command Name	Notes	Parameters								
Get Motor Status	Get the status of the valve.	<u>Name</u>		<u>Type</u>		<u>Notes</u>				
		Valve		String		Will be one of these values. 'R' for Moving, 'L' for Load position, 'I' for Inject position.				
Get Valve Position	Gets the valve position. 'R' for Moving, 'L' for Load position, 'I' for Inject position.	<u>Name</u>		<u>Type</u>		<u>Notes</u>				
		Valve Position		String						
Identify	The PROM instrument name and version.	<u>Name</u>		<u>Type</u>		<u>Notes</u>				
		Identity		String						
Immediate	Legacy.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>			
		Command	String	%		-				
		<u>Name</u>	<u>Type</u>	<u>Notes</u>						
		Response	String							
Move Valve to Inject	Turns valve to injection position.	<u>Name</u>								
		Valve								
Move Valve to Load	Turns valve to load position.	<u>Name</u>								
		Valve								
Valve Position	Moves the valve to specified position. 'L' for Load position, 'I' for Inject position.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>			
		Valve Position	String	L		-				
		<u>Name</u>								
		Valve								
Wait For Valve	Wait for valve motor to come to load position.	<u>Name</u>								
		Valve								

406 Single Syringe Pump

Device Name	406 Single Syringe Pump
Version	1.0.7.0
Unit ID	7

Status Command	Frequency
Get Error	Sent once per command
Get Pressure	Always on
Get Syringe Info	Always on

Command Name	Notes	Parameters					
Aspirate	Flip valve to position specified and then aspirate with the specified flow rate.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Valve Position	String	N		-	'N' for needle or 'R' for reservoir position.
		Volume	Number	0	μL	-	Volume in μL to aspirate.
		Flow Rate	Number	10	mL/min	-	Flow Rate at which to aspirate in mL/min
		<u>Name</u>					
		Syringe					
Buffered	Legacy.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Command	String			-	
Clear Error	Clears error code.						

Command Name	Notes	Parameters					
Dispense	Flip valve to specified position and then dispense at specified flow rate.	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Valve Position	String	N		-	'N' for needle or 'R' for reservoir position.
		Volume	Number	0	µL	-	Volume to dispense in µL.
		Flow Rate	Number	10	mL/min	-	Flow rate at which to dispense in mL/min.
		<u>Name</u>					
		Syringe					
Get Error	Gets error code. See table.	<u>Name</u>		<u>Type</u>		<u>Notes</u>	
		Error		String			
Get Motor Status	Gets the status of the syringe.	<u>Name</u>		<u>Type</u>		<u>Notes</u>	
		Syringe		String		Will be one of these values. 'U' for unpowered, 'P' for parked, 'R' for running, 'E' for Error, 'X' for missing.	
Get Pressure	Gets current pressure value with 2 decimal places.	<u>Name</u>		<u>Type</u>		<u>Notes</u>	
		Pressure		Number			
Get Syringe Info	Gets the valve position and Syringe contents volume in µL or '?' if unknown (Homed).	<u>Name</u>		<u>Type</u>		<u>Notes</u>	
		Valve Position		String		N, R, or G for 3 position valve, N or R for 2 position valve, T if configured for a 'tee', X for missing, E if an invalid configuration	
		Volume		Number			
Halt Syringe	Syringe operation is halted.	<u>Name</u>					
		Syringe					

Command Name	Notes	Parameters					
Home	Initialized Valve & Syringe. NOTE: Some liquid will typically come from the probe during initializing, so ensure that the probes are in a suitable location before initializing.	<u>Name</u> Syringe					
Identify	The PROM instrument name and version	<u>Name</u>		<u>Type</u>	<u>Notes</u>		
		Identity		String			
Immediate	Legacy	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Command	String	%		-	
		<u>Name</u>		<u>Type</u>	<u>Notes</u>		
		Response		String			
Valve Position	Gets the valve position of the pump. N, R, or G for 3 position valve, N or R for 2 position valve, T if configured for a 'tee', X for missing, E if an invalid configuration	<u>Name</u>	<u>Type</u>	<u>Default</u>	<u>Units</u>	<u>Range</u>	<u>Notes</u>
		Valve Position		String	N	-	
		<u>Name</u>					
		Syringe					
Wait For Syringe	Wait for syringe to complete the current operation.	<u>Name</u>					
		Syringe					

Non-Volatile Memory Map

GX-27X

Address	Meaning	Units	Decimal Places	Default
0	Reserved			
1	Reserved			
2	Controls the output contact and +24V output operation. Bit format is an 8 bit field, (0 to 0FF in hexadecimal). The 8 bit field is viewed as four sub-fields, each containing two bits. Each two bit sub-field controls one contact. bits 8 & 7 control CC1 bits 6 & 5 control CC2 bits 4 & 3 control 24V1 bits 2 & 1 control 24V2 For each sub-field valid values are: 0 = no action 1 = contact closes (or power on) 2 = contact opens (or power off)	x	0	10101010b
3	Park position X coordinate	mm	3	0
4	Park position Y coordinate	mm	3	0
5	Clamp height of the Z drive in mm The minimum value is 110 mm	mm	3	125.000
6	Reserved			
7	Liquid sensitivity	%	2	1
8 to 24	Reserved			
25	X offset	mm	3	0
26	Y offset	mm	3	0
27	Z offset	mm	3	0
28 to 31	Reserved			

GX Solvent System

Address	Meaning	Units	Decimal Places	Default
0	Waste port number		0	1
1	Transfer tubing volume	μL	0	1000
2 to 29	Reserved			
30	Service life – pump head revolutions		0	
31	Service life – valve actuations		0	

GX Direct Injection Module

Address	Meaning	Units	Decimal Places	Default
0 to 3	Reserved			
4	Valve switch speed 9600 yields approximately 200 msec valve switch time for 1/16" valve 4800 yields approximately 300 msec valve switch time for 1/8" valve	uSteps/second	0	4800
5 to 30	Reserved			
31	Service life – valve actuations		0	0

406 Single Syringe Pump

Address	Meaning	Units	Decimal Places	Default
0	Language	Decimal separator: 0 for '.' 1 for ','	0	0
1	Reserved			
2	Left valve positions	0 = "Tee", 2 = Std, 3 = ASPEC	0	3
3	Right valve positions	0 = "Tee", 2 = Std, 3 = ASPEC	0	3

Address	Meaning	Units	Decimal Places	Default
4	Left syringe size	Contains syringe size in μL as an integer, with 0 = missing. Other permitted values: 100 = 100 μL syringe 250 = 250 μL syringe 500 = 500 μL syringe 1000 = 1 mL syringe 5000 = 5 mL syringe 10000 = 10 mL syringe 25000 = 25 mL syringe	0	10000
5	Right syringe size	Contains syringe size in μL as an integer, with 0 = missing. Other permitted values: 100 = 100 μL syringe 250 = 250 μL syringe 500 = 500 μL syringe 1000 = 1 mL syringe 5000 = 5 mL syringe 10000 = 10 mL syringe 25000 = 25 mL syringe	0	10000
6	Left Backlash	μSteps	0	100
7	Right Backlash	μSteps	0	100
8	Left pressure sensor offset	Bar	2	0.00
9	Right pressure sensor offset	Bar	2	0.00
10	Left pressure sensor scale	PPM	0	0
11	Right pressure sensor scale	PPM	0	0
12..31	Reserved			

Error Messages

GX-27X

Error Number	Error Text
0	no error
10	Unknown command
11	Invalid NV-RAM address
12	Safety contact closed
13	Invalid command parameter
14	S buffer overflow
15	S command while unhomed
19	X encoder error
20	Y encoder error
21	X homing error
22	Y homing error
23	XY target out of range
24	XY speed invalid
25	X stall or jam
26	Y stall or jam
27	XY move while unhomed
28	XY move while busy
29	Park location invalid
31	Z homing error
32	Z target out of range
33	Z speed invalid
34	Z stall or jam
35	Z move while unhomed
36	Z move while busy

GX Solvent System

<i>Error Number</i>	<i>Error Text</i>
1	Bad command
2	Invalid NV address
3	Error during homing
4	Setting valve position prior to home
5	Motion request while still moving
6	Valve encoder disagreement
7	Aspirate more than transfer volume
8	Dispense from waste
9	Aspirate to non-waste port
10	Flow rate out of range
11	Invalid valve position

GX Direct Injection Module

<i>Error Number</i>	<i>Error Text</i>
0	No error
1	Unknown command
2	Invalid NV-RAM address specified
3	Previous move not complete
4	Invalid position requested

406 Single Syringe Pump

Error Number	Error Text
0	No Error
1	Left valve absent
2	Right valve absent
3	Left syringe absent
4	Right syringe absent
5	Left valve position invalid
6	Right valve position invalid
7	Left volume invalid
8	Right volume invalid
9	Left syringe jammed
10	Right syringe jammed
11	Left flow rate invalid
12	Right flow rate invalid
13	Left syringe size invalid
14	Right syringe size invalid
15	Left unit unhomed
16	Right unit unhomed
17	Left side busy
18	Right side busy
20	Unknown command
21	Invalid NV-RAM address
22	Left valve type invalid
23	Right valve type invalid